Software Engineering Department
Braude College of Engineering
Final Project – Phase A (61998)

Compressing the Earthquake Transformer via Knowledge Distillation

Enhancing Efficiency in Seismic Analysis

CODE: 26-2-R-3

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Abstract

Recent advances in deep learning have significantly improved the accuracy of seismic signal analysis and earthquake detection systems. In particular, Transformer based architectures, such as the Earthquake Transformer, have demonstrated strong performance in identifying seismic events by utilizing attention mechanisms to capture complex relationships in seismic waveform signals. However, despite their high accuracy, deploying these models in large scale seismic monitoring systems remains challenging due to real time processing requirements and inference latency constraints. As seismic networks continuously generate large volumes of waveform data, efficient inference becomes essential for practical deployment on edge devices and resource constrained platforms.

This project investigates the application of knowledge distillation techniques to compress and optimize Transformer based seismic models while preserving their predictive capabilities. Inspired by successful model compression approaches in natural language processing, such as DistilBERT, the research explores whether similar strategies can be adapted effectively to the seismic domain.

The project focuses on developing a smaller and more efficient student model derived from the original Earthquake Transformer architecture. Different compression and distillation approaches will be evaluated in terms of model size, inference speed, and predictive performance. The ultimate goal is to enable efficient real time seismic analysis on resource constrained systems while maintaining reliable earthquake detection capabilities.

The expected contribution of this work is a practical methodology for compressing seismic deep learning models, demonstrating how a smaller student architecture can support faster and more efficient earthquake detection and phase picking while remaining close to the performance of the original EQTransformer.

Introduction

Background and Motivation

Earthquake detection and seismic monitoring systems play a critical role in reducing the impact of natural disasters and improving early warning capabilities. Modern seismic networks continuously generate large volumes of waveform data, creating a growing need for accurate and efficient automated analysis systems. In recent years, deep learning approaches have significantly improved seismic event detection and phase picking compared to traditional signal processing techniques.

Among these approaches, Transformer based architectures have shown particularly strong performance due to their ability to capture long range temporal dependencies using attention mechanisms. One of the most notable models in this field is the Earthquake Transformer, developed by researchers at Stanford University. The model analyzes seismic waveform signals and identifies important seismic phases such as P-waves and S-waves to detect earthquakes and estimate their characteristics. Despite its effectiveness and high detection accuracy, deploying EQTransformer in large scale seismic monitoring systems remains challenging. Modern seismic networks continuously process large volumes of waveform data generated by numerous monitoring stations. As a result, inference latency and computational efficiency become critical factors, particularly in real time monitoring environments and low cost edge devices.

Limitations of Existing Models

Although EQTransformer achieves state of the art performance in earthquake detection and seismic phase picking, its deployment still involves several practical limitations. The model includes multiple processing components, including convolutional layers, recurrent layers, attention mechanisms, and separate output branches for detection and phase picking. This structure increases the computational cost of continuous inference, especially when predictions must be produced repeatedly over long waveform streams.

In addition, real time seismic monitoring systems often operate under strict latency and hardware constraints. Even when a model is not extremely large in terms of parameter count, repeated inference across many stations can lead to significant cumulative computation. This makes efficiency, memory usage, and inference speed important factors in practical deployment.

Therefore, there is a need for compressed seismic models that preserve the predictive strength of EQTransformer while reducing computational overhead and improving suitability for resource constrained environments.

Knowledge Distillation for Seismic Models

At the same time, the field of natural language processing introduced highly successful model compression techniques designed to reduce the size and complexity of large Transformer models. One of the most prominent examples is DistilBERT, which applies knowledge distillation methods to create a smaller and faster student model while preserving most of the original model's predictive performance.

Inspired by the success of DistilBERT, this project investigates whether similar compression techniques can be applied to Transformer based seismic models. Specifically, the research focuses on compressing the EQTransformer architecture using knowledge distillation in order to

reduce computational requirements while preserving reliable earthquake detection and seismic phase picking performance.
The project explores different architectural reduction strategies in order to identify which layers can be removed while preserving the model's structural integrity and predictive capabilities. After constructing a smaller architecture, knowledge distillation techniques are applied so that the compressed student model learns directly from the original teacher model.

Research Objectives

The main objective of this project is to design and evaluate compressed student versions of EQTransformer using knowledge distillation. The research aims to reduce the computational requirements of the original model while preserving its ability to detect earthquake events and pick P-wave and S-wave arrivals.
The project will compare several student architectures according to model size, number of parameters, memory consumption, inference time, and detection and phase picking performance. In addition, the feasibility of using the compressed model in low resource environments, such as edge devices, will be examined.
The final goal is to identify an effective trade off between efficiency and predictive performance, and to determine whether knowledge distillation can be adapted successfully from Transformer based language models to seismic waveform analysis.

Literature Review

Seismic Signal Analysis

Recent advances in deep learning have significantly transformed the field of seismology, particularly in earthquake detection and seismic phase picking. Traditional approaches relied heavily on manual analysis and signal processing techniques, which often struggle to handle the growing volume of seismic data generated by modern monitoring networks. As a result, researchers have increasingly adopted deep learning methods to automate and improve seismic event detection.

Earthquake Detection and Phase Picking

Earthquake detection and phase picking are fundamental tasks in modern seismology. Seismic waves generated by earthquakes propagate through the Earth and are recorded by seismic stations. Among these waves, P-waves (Primary waves) and S-waves (Secondary waves) are of particular importance because their arrival times are used to estimate earthquake locations and magnitudes.
Traditionally, phase picking relied on signal processing techniques and manual expert analysis. Although these methods have been successfully used for decades, they often struggle when dealing with noisy signals and the large volumes of data produced by modern seismic networks. Recent advances in deep learning have enabled automatic phase picking systems that outperform traditional approaches. Neural networks can learn complex patterns directly from waveform data and provide accurate detection and arrival time estimation, making them highly suitable for real time seismic monitoring applications.

Transformer Architectures

Transformer architectures were originally introduced in the field of natural language processing through the work "Attention Is All You Need" by Vaswani et al. (2017). Unlike recurrent neural networks, Transformers rely on self attention mechanisms that allow the model to capture long range dependencies within sequential data.
The self attention mechanism enables the model to focus on the most relevant parts of the input sequence when generating predictions. As a result, Transformer models have achieved state of the art performance across numerous domains, including language processing, computer vision, speech recognition, and scientific data analysis.
Their ability to model complex temporal relationships makes transformer architectures particularly attractive for seismic signal analysis, where important information may be distributed across long waveform sequences.

Earthquake Transformer

One of the most influential contributions in this field is the Earthquake Transformer (EQTransformer), introduced by Mousavi et al. (2020). The model combines an encoder decoder architecture with attention mechanisms inspired by Transformer networks. By analyzing seismic waveform signals, EQTransformer performs simultaneous earthquake detection and P-wave and S-wave phase picking with high accuracy. Experimental results demonstrated that the model outperformed several existing deep learning approaches such as PhaseNet (Zhu & Beroza, 2019) while maintaining robustness across different seismic datasets.

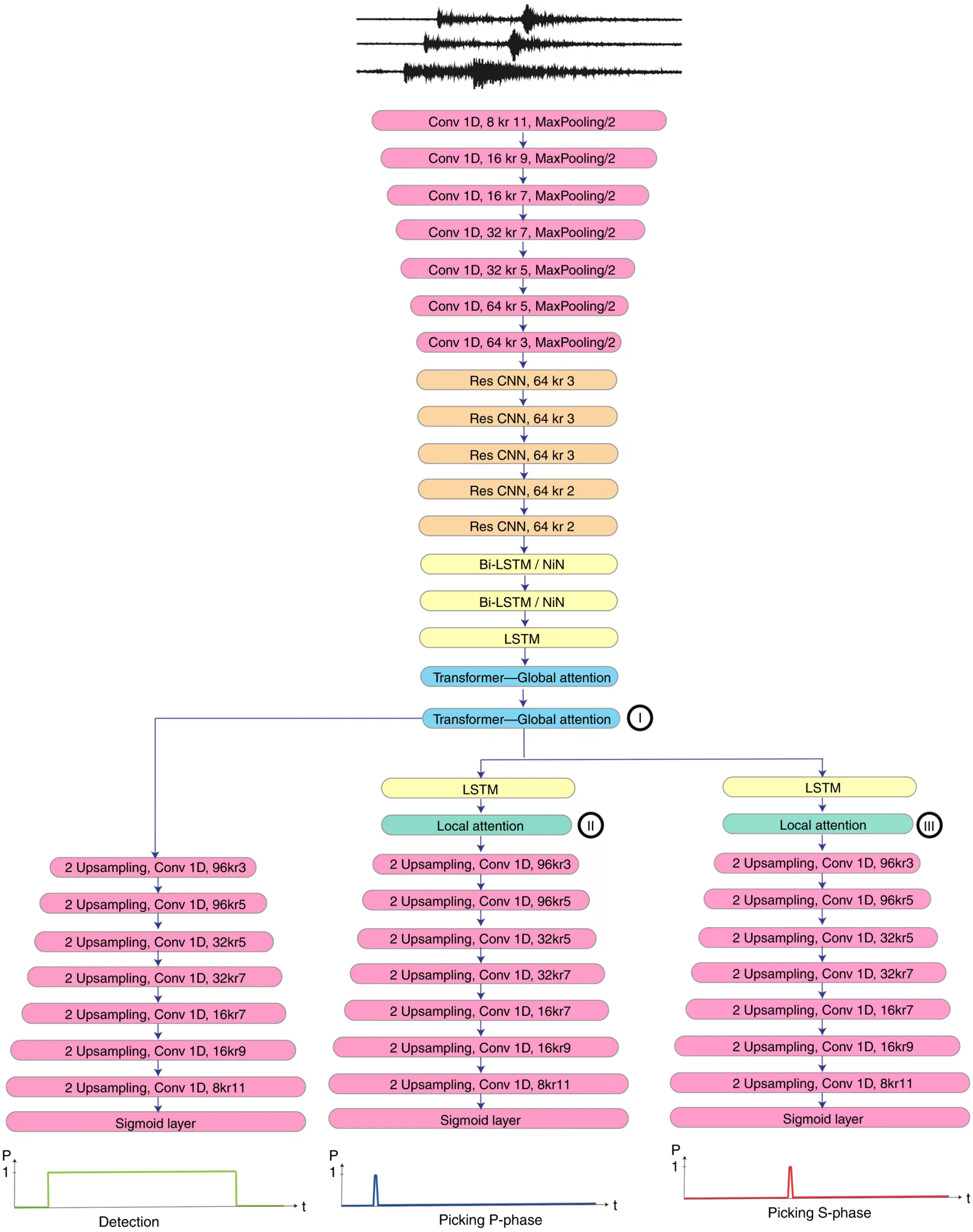

Figure 1: Architecture of the Earthquake Transformer model.
Note. Source: Mousavi et al. (2020).

As shown in Figure 1, EQTransformer employs a hybrid architecture consisting of convolutional layers, residual CNN blocks, recurrent networks, and Transformer based attention mechanisms. The model simultaneously performs earthquake detection, P-wave picking, and S-wave picking through three dedicated output branches.

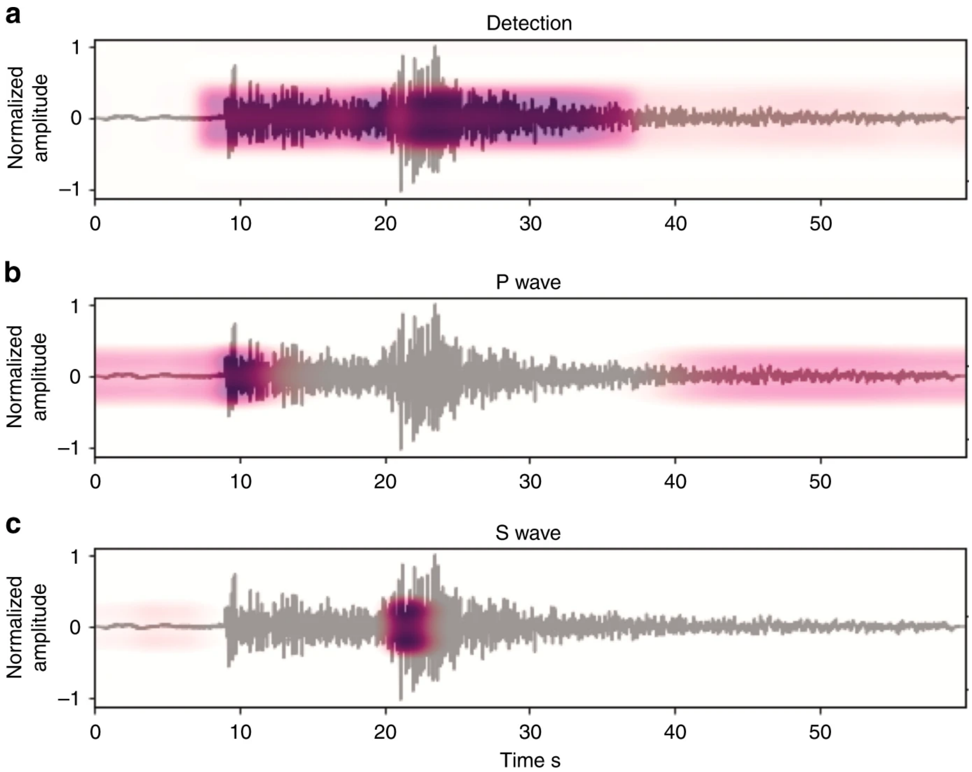

Figure 2: Example of earthquake detection and seismic phase picking performed by EQTransformer.

Note. Source: Mousavi et al. (2020).

As shown in Figure 2, EQTransformer simultaneously predicts earthquake detection probabilities and identifies the arrival times of P-waves and S-waves directly from seismic waveform signals. This multitask learning framework enables accurate characterization of seismic events using a single model.
To evaluate the effectiveness of the model, Mousavi et al. conducted extensive experiments on multiple seismic datasets. The results demonstrated highly accurate P-wave and S-wave phase picking performance, with very low timing errors and strong generalization capabilities across different seismic events.

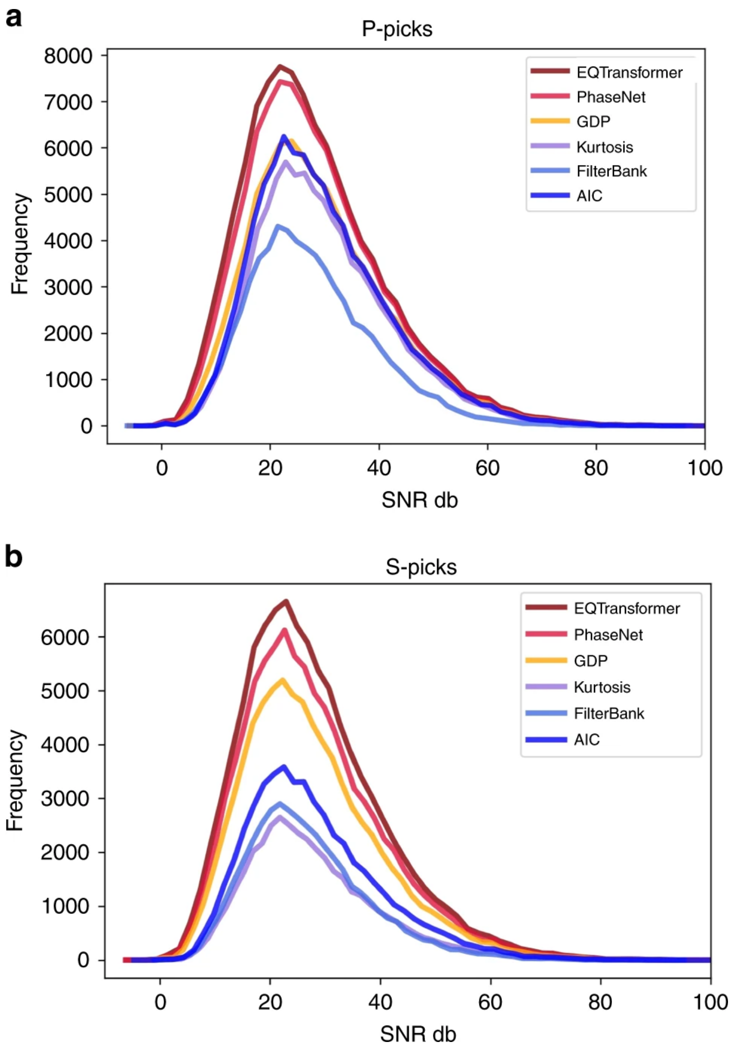

Figure 3: Performance of EQTransformer on seismic phase picking.
Note. Source: Mousavi et al. (2020).

As shown in Figure 3, EQTransformer produces a larger number of reliable P-wave and S-wave picks across a broad range of signal to noise ratios compared with the evaluated baseline methods. These results demonstrate its robustness in processing seismic signals under varying noise conditions.

Despite its strong performance, EQTransformer can be computationally demanding in practical deployment scenarios. Although the model is relatively compact, performing continuous inference across large seismic networks requires substantial cumulative computation, which restricts its deployment in real time and resource constrained environments such as edge devices and distributed seismic monitoring stations.

Knowledge Distillation

Model compression techniques have emerged as an effective solution for reducing the computational requirements of deep neural networks. Among these techniques, knowledge distillation has become one of the most widely adopted approaches. Originally proposed by Hinton et al. (2015), knowledge distillation transfers knowledge from a large pretrained teacher model to a smaller student model. Rather than learning directly from labeled data alone, the student model

learns to mimic the behavior of the teacher, allowing it to achieve competitive performance with substantially fewer parameters.
Knowledge Distillation has become increasingly important as deep learning models continue to grow in size and complexity. Large neural networks often achieve superior performance, but their computational requirements make deployment difficult in practical environments. Knowledge Distillation addresses this challenge by enabling smaller models to inherit the knowledge learned by larger models.
The distillation process typically involves minimizing the difference between the output probability distributions of the teacher and student models. Rather than relying solely on ground truth labels, the student model learns from the soft predictions generated by the teacher model. These predictions contain additional information about class relationships and decision boundaries, which can improve the learning process.
As a result, knowledge distillation has been successfully applied in various domains, including computer vision, natural language processing, speech recognition, and recommendation systems. The technique has demonstrated the ability to significantly reduce computational requirements while preserving a large portion of the original model's performance. These promising results motivate the investigation of knowledge distillation techniques for seismic signal analysis models, particularly Transformer based architectures such as EQTransformer.

DistilBERT

A notable application of knowledge distillation is DistilBERT, introduced by Sanh et al. (2019). DistilBERT demonstrated that the BERT language model (Devlin et al., 2019) could be reduced by approximately 40% while retaining around 97% of the original model's performance. This work showed that substantial reductions in model size and computational cost are possible without significant degradation in accuracy.
The success of DistilBERT demonstrated that substantial reductions in model complexity can be achieved by training a smaller student model under the guidance of a pretrained teacher model. By leveraging the knowledge already captured by the teacher model, the student model is able to learn more efficiently and converge faster during training.
The principles introduced by DistilBERT have inspired numerous subsequent studies on model compression and efficient transformer architectures. These works suggest that knowledge distillation is not limited to language models and may be applicable to a wide range of Transformer based systems operating in different domains.
This observation serves as the primary motivation for the current research. Since EQTransformer is also based on transformer architectures, this project investigates whether the compression strategies successfully applied in DistilBERT can be adapted to seismic signal analysis while maintaining reliable earthquake detection and phase picking performance.

Knowledge Distillation Process

Knowledge Distillation is based on a teacher student learning framework. In this approach, a large pretrained model serves as the teacher, while a smaller and more efficient model serves as the student. Instead of learning only from the original ground truth labels, the student model is trained to imitate the behavior of the teacher model.
The teacher model generates probability distributions over the output classes, often referred to as soft labels. Unlike hard labels, which contain only the correct class, soft labels provide additional information regarding the confidence of the teacher model and the relationships between

different classes. This information enables the student model to learn richer representations and improve its generalization capabilities.
To further enhance the transfer of knowledge, a technique known as temperature scaling is commonly applied in classification based distillation. A temperature parameter (T) is introduced to soften the output probability distributions produced by the teacher model. Higher temperature values generate smoother probability distributions, allowing the student model to capture more information from the teacher's predictions during training.
However, temperature scaling is mainly relevant to classification tasks with discrete class probabilities. In this project, EQTransformer produces continuous probability sequences for earthquake detection, P-wave picking, and S-wave picking. Therefore, the proposed distillation approach focuses on matching the teacher and student output curves using regression based objectives, rather than relying on temperature scaled class probabilities.
The training objective typically combines two loss functions: the traditional student loss, which measures the difference between the student predictions and the ground truth labels, and the distillation loss, which measures the difference between the student and teacher outputs. The combined loss function can be expressed as:

$$L = \alpha L_{KD} + (1 - \alpha) L_{Student}$$

where L_{KD} represents the distillation loss, $L_{Student}$ represents the standard training loss, and α controls the balance between the two objectives. In the proposed implementation, the student loss will correspond to the original EQTransformer training objective, while the distillation loss will be computed as the Mean Squared Error (MSE) between the output probability curves of the teacher and student models.
In the proposed research, the teacher model transfers its knowledge through the probability curves generated for each seismic waveform. The student model is trained to reproduce these curves while also learning from the original seismic labels, with similarity measured using regression based objectives such as Mean Squared Error (MSE). This allows the student model to learn the temporal patterns captured by the original EQTransformer.
Specifically, the original EQTransformer model will serve as the teacher model, while a compressed architecture with fewer layers and parameters will serve as the student model. This approach is expected to preserve a large portion of the teacher model's performance while significantly reducing computational requirements.

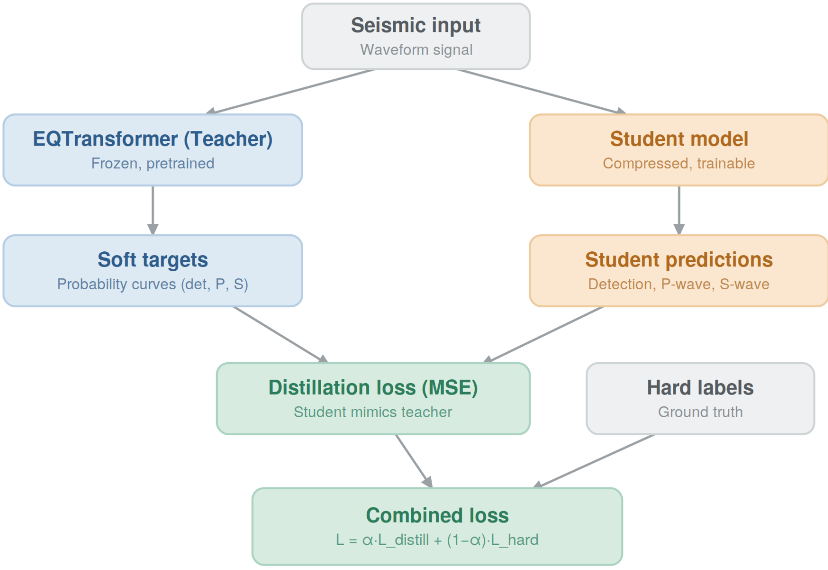

Figure 4: Knowledge Distillation process used in the proposed research.
Note. Source: Authors' illustration.

As shown in Figure 4, the proposed knowledge distillation framework consists of a pretrained EQTransformer teacher model and a compressed student model. Seismic waveform recordings are provided as input to both models. The teacher model generates probability curves for earthquake detection, P-wave picking, and S-wave picking, which serve as soft targets for the student model. The student is trained to reproduce these outputs while simultaneously learning from the original seismic labels through a combined loss function. This process enables the student model to inherit the knowledge of the teacher while maintaining a significantly smaller architecture and lower computational requirements.

Research Gap

Motivated by the success of knowledge distillation in natural language processing, this project investigates whether similar compression techniques can be applied to Transformer based seismic models. In particular, the research explores the compression of EQTransformer through controlled architectural reduction and teacher student training.

Although knowledge distillation has been studied extensively in natural language processing and computer vision, its application to Transformer based seismic models remains limited. It is therefore still unclear whether a substantially smaller EQTransformer architecture can preserve both earthquake detection and phase picking performance through knowledge distillation. This project addresses this research gap by combining architectural reduction with output level distillation and evaluating the resulting student models on both in domain and cross domain seismic data.

The objective is to develop a lightweight model that reduces computational requirements while preserving reliable earthquake detection and phase picking performance, thereby improving its suitability for low resource platforms and edge computing environments.

Project Description

The final outcome of the project will be a compressed EQTransformer model trained using knowledge distillation, designed to preserve earthquake detection and phase-picking performance while reducing computational requirements.

Functional Requirements

The main objective of this project is to develop a compressed version of the Earthquake Transformer (EQTransformer) while preserving its earthquake detection and seismic phase picking capabilities.
The system should be able to:

- Detect earthquake events from seismic waveform data.
- Perform automatic P-wave phase picking.
- Perform automatic S-wave phase picking.
- Train a compressed student model using knowledge distillation.
- Evaluate the performance of the student model against the original EQTransformer model.
- Perform evaluation on both STEAD and K-NET datasets.

Non Functional Requirements

The following non functional requirements define the main quality attributes of the proposed system. Each requirement is mapped to a relevant category, such as performance, reliability, efficiency, portability, or testability.

Table 1: Non functional requirements and their classifications.

ID	Non Functional Requirement	Classification
NFR1	Reduce the number of model parameters compared to the original EQTransformer.	Efficiency / Capacity
NFR2	Reduce memory consumption to support execution on limited resource hardware.	Efficiency / Resource Utilization
NFR3	Reduce inference time for faster seismic waveform processing.	Performance / Latency
NFR4	Maintain earthquake detection accuracy close to the original teacher model.	Reliability / Accuracy
NFR5	Maintain P-wave and S-wave picking performance.	Reliability / Accuracy
NFR6	Support deployment on resource constrained edge devices.	Portability / Resource Constraints
NFR7	Preserve an experimental pipeline that can be evaluated on both STEAD and K-NET datasets.	Testability / Portability

Requirements Collection

The project requirements were derived from an extensive review of scientific literature and technical documentation. The primary sources include the original EQTransformer and DistilBERT papers, which provide the theoretical foundations for seismic signal analysis, Transformer based modeling, and knowledge distillation. Additional requirements were derived from the STEAD and K-NET dataset documentation, as well as from evaluation methodologies and performance metrics commonly used in seismic phase picking and model compression research.

Datasets

The project utilizes two primary seismic datasets.
The first dataset is STEAD (Stanford Earthquake Dataset; Mousavi et al., 2019), which contains hundreds of thousands of labeled seismic waveform recordings collected from multiple seismic stations. STEAD serves as the primary dataset for training and evaluating both the teacher and student models.
The second dataset is K-NET, a Japanese strong motion seismic dataset widely used in earthquake research (Okada et al., 2004). K-NET is employed as a cross domain evaluation benchmark to assess the generalization capabilities of the compressed model on unseen seismic data collected under different geographical and instrumental conditions.
Using both datasets enables evaluation of not only model accuracy but also robustness and transferability across different seismic environments.

Research Methodology

The research process consists of several stages.
First, the original EQTransformer architecture will be studied and reproduced in order to establish a baseline for comparison. Next, different architectural reduction strategies will be investigated, including reducing encoder depth, removing selected layers, and decreasing the number of filters.
Several compressed student architectures will be investigated. The planned modifications include reducing the number of Transformer blocks, decreasing the number of convolutional filters, simplifying the recurrent layers, and combining these modifications into a single compressed architecture. The final student architecture will be selected based on the experimental results.
Before designing the compressed student architectures, an initial validation experiment was conducted using the original pretrained EQTransformer model. The model was executed on the official sample dataset provided with the EQTransformer repository. This dataset contains 100 labeled seismic waveform recordings originating from the STEAD dataset. Detection, P-wave, and S-wave probability thresholds were all set to 0.3 based on the experimental configuration used in this study.
The purpose of this experiment was to verify the correct operation of the original model and to establish a quantitative baseline for the subsequent compression experiments. Inspection of the pretrained EQTransformer architecture showed that the model contains 373,495 parameters, including 371,639 trainable parameters, with an approximate storage size of 1.4 MB. These characteristics serve as the reference point for assessing the effectiveness of the proposed compression strategies and measuring the reduction in model complexity achieved by the distilled student models.

The results showed that EQTransformer successfully identified 99 out of 100 earthquake events, while one waveform was not classified as a seismic event. For a representative detection example, the model produced a detection probability of 0.99, a P-wave probability of 0.96, and an S-wave probability of 0.68. The lower S-wave confidence in this example may be related to the greater difficulty of identifying the S-wave arrival within a more complex portion of the waveform.
The purpose of this preliminary experiment was to verify the correct execution of the original EQTransformer and establish an initial reference point for the planned compression experiment.

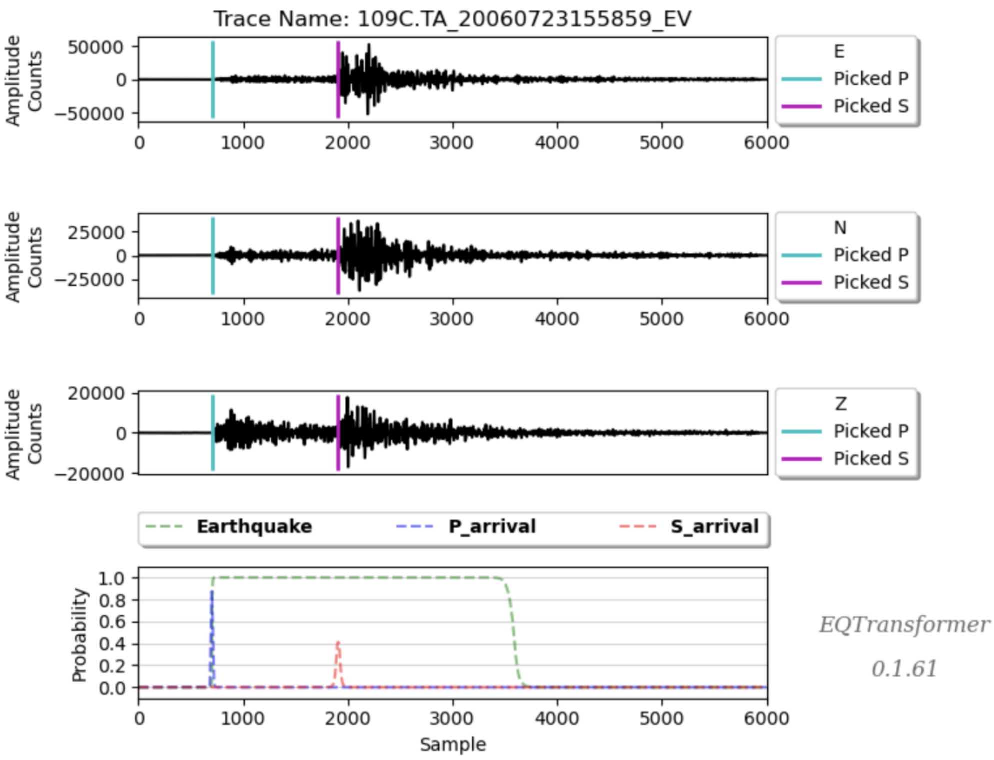

Figure 5: Example results obtained from running the pretrained EQTransformer model on a sample dataset of 100 seismic recordings.

Note. Source: Authors' experiment.

Figure 5 presents the output generated by the original EQTransformer model during the preliminary validation experiment. The results confirm the model's ability to accurately detect earthquake events and identify seismic phase arrivals prior to the compression stage.
After constructing a smaller student architecture, knowledge distillation techniques will be applied. The student model will learn from the predictions generated by the pretrained teacher model. To evaluate the effectiveness of knowledge distillation, the performance of the distilled student model will also be compared with the same compressed architecture trained directly on the original dataset without knowledge distillation.
The distilled model will be trained and evaluated using the STEAD dataset. Finally, both the teacher and student models will be evaluated on the K-NET dataset in order to assess their generalization capabilities on unseen seismic data. The STEAD dataset will be divided into training, validation, and testing subsets. The same data split will be used throughout all experiments to ensure a fair comparison between the teacher model and all proposed student architectures.

Several student architectures will be designed and evaluated throughout the research process. Each architecture will differ in the number of layers, encoder depth, and parameter count. A conceptual comparison between the original teacher model and a potential compressed student architecture is presented in Figure 6.

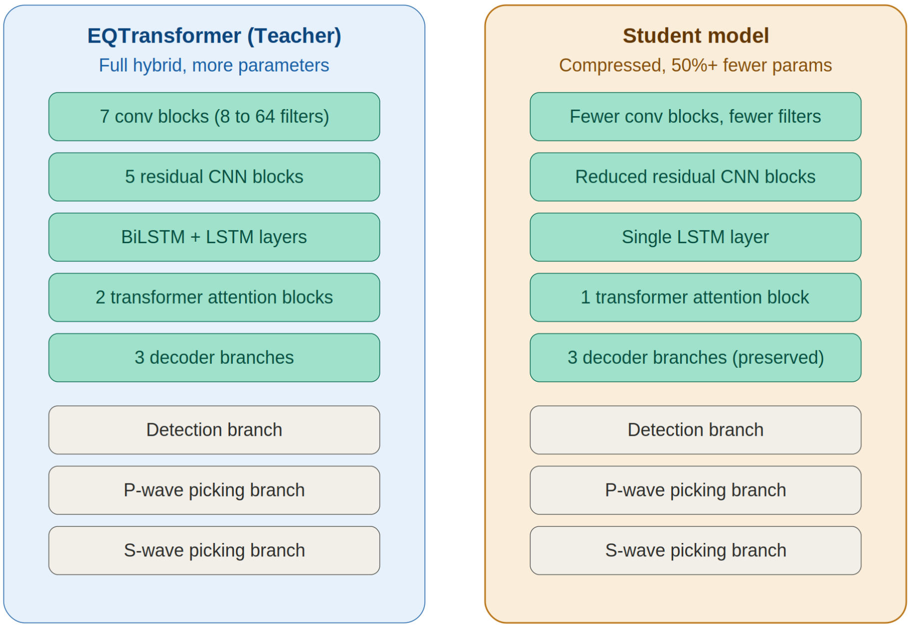

Figure 6: Conceptual comparison between the original EQTransformer teacher model and a compressed student architecture.

Note. Source: Authors' illustration.

Figure 6 presents a conceptual illustration of the proposed compression strategy. The student architecture contains fewer layers and parameters than the original teacher model while preserving the overall structure required for earthquake detection and seismic phase picking. The exact architecture of the student model will be determined experimentally during the implementation phase based on the results of the compression and distillation experiments.

An ablation study will also be conducted in order to evaluate the contribution of different architectural components. Several student architectures will be created by progressively reducing encoder depth, convolutional filters, residual blocks, and recurrent layers. The resulting models will be compared in terms of earthquake detection performance, phase picking accuracy, model size, memory consumption, and inference speed. This analysis will provide a quantitative justification for the architectural decisions adopted in the final compressed model.

This experimental approach will allow the identification of the most effective trade off between computational efficiency and predictive performance.

The experiments will be conducted in a controlled environment to ensure fair comparisons between models. All models will be trained using identical datasets and evaluated using the same performance metrics. The resulting data will be analyzed to determine the effectiveness of different compression strategies.

Expected Challenges

Several challenges are expected during the project:

- **Maintaining Earthquake Detection and Phase Picking Accuracy**
  Reducing the size of the model may negatively affect earthquake detection performance and the accuracy of P-wave and S-wave arrival predictions. To address this challenge, the compressed student models will be continuously compared against the original EQ-Transformer using detection and phase picking metrics.

- **Selecting Layers for Compression**
  Identifying which layers can be removed without significantly degrading performance is a complex task. Different architectural reduction strategies will be investigated in order to determine the most suitable compression configuration.

- **Balancing Efficiency and Performance**
  A smaller model is expected to be faster and more efficient, but excessive compression may reduce prediction quality. The project will therefore explore several student architectures to identify the optimal trade off between model size and performance.

- **Preserving Generalization Capabilities**
  A compressed model may perform well on the training dataset but fail to generalize to unseen seismic events. To mitigate this risk, the student model will be evaluated on both the STEAD and K-NET datasets.

- **Ensuring Effective Knowledge Transfer**
  The student model may not fully capture the knowledge learned by the teacher model. Different distillation settings and hyperparameters will be examined in order to maximize the effectiveness of the knowledge transfer process.

Tools and Technologies

The following tools and technologies will be used throughout the project:

- Python
- TensorFlow / Keras
- EQTransformer
- Knowledge Distillation
- STEAD Dataset
- K-NET Dataset
- Google Colab
- Git
- GitHub

Proposed Solution

The proposed solution is to develop a compressed student version of EQTransformer using knowledge distillation. Instead of replacing the original model with an unrelated architecture, the student model is designed to preserve the main detection and phase picking functionality of the teacher while using fewer computational resources.

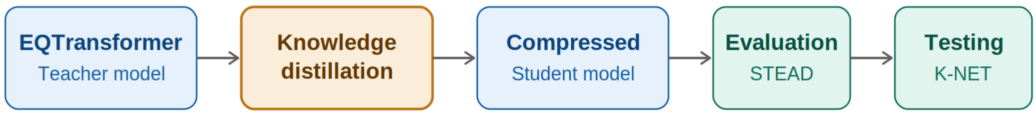

Figure 7: Proposed methodology for compressing EQTransformer using Knowledge Distillation.

Note. Source: Authors' illustration.

As illustrated in Figure 7, the workflow begins with the pretrained EQTransformer teacher model. The teacher generates detection, P-wave, and S-wave probability curves, which are used as soft targets for training the compressed student model. The trained student is then evaluated on STEAD and tested on K-NET to assess both in domain performance and cross domain generalization.

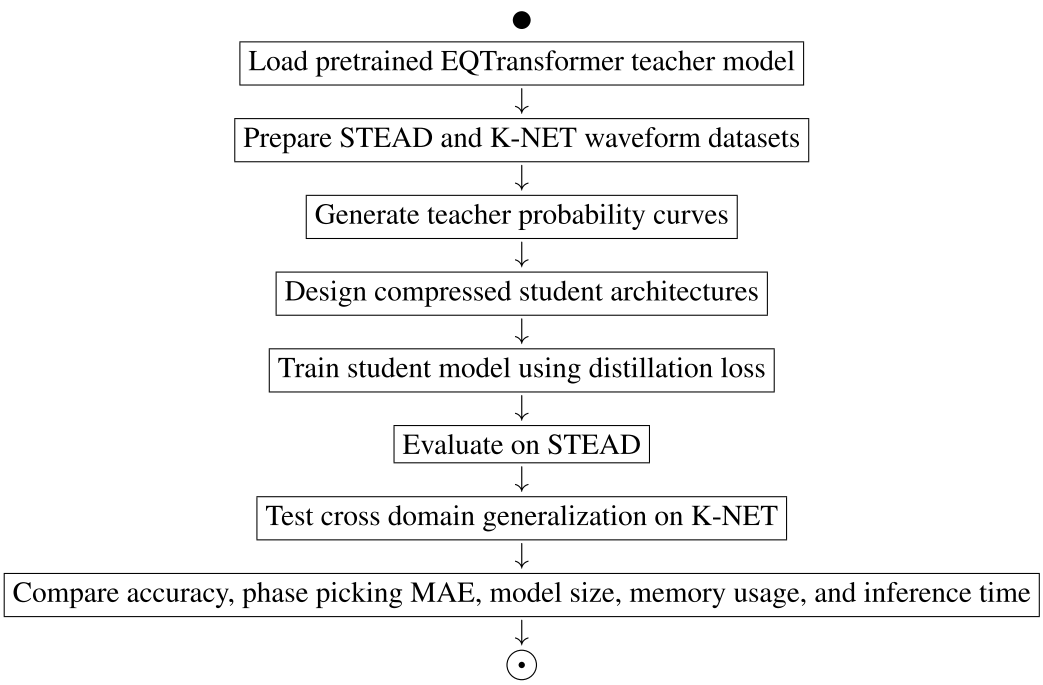

Figure 8: Activity diagram of the proposed compression and evaluation pipeline.

Figure 8 presents the main activity flow of the proposed research pipeline. The process starts from the pretrained teacher model, continues through student model training using knowledge distillation, and ends with quantitative evaluation on both in domain and cross domain datasets.

This solution focuses on preserving the practical seismic capabilities of EQTransformer while improving efficiency in terms of model size, memory consumption, and inference time.

Success Metrics

The success of the project will be evaluated using the following quantitative metrics:

- Reduce the number of model parameters by at least 50%.
- Reduce inference time compared to the original EQTransformer.
- Retain at least 95% of the teacher model's detection F1-score while maintaining phase picking MAE close to that of the teacher model.
- Preserve reliable earthquake detection capabilities.
- Preserve reliable P-wave and S-wave phase picking performance.
- Demonstrate successful operation on both STEAD and K-NET datasets.

Testing Process

The evaluation process will include multiple experiments and comparisons between the teacher and student models.
The models will be tested on the STEAD dataset and evaluated using common machine learning metrics such as accuracy, precision, recall, and F1-score. Additional experiments will measure model size, memory consumption, and inference speed.
To evaluate generalization capabilities, both models will also be tested on the K-NET dataset. The results will then be compared in order to determine the effectiveness of the proposed compression approach.
The evaluation framework will include both quantitative and qualitative analysis. Quantitative evaluation will focus on prediction accuracy, precision, recall, F1-score, phase picking MAE, and inference latency. These metrics will provide insight into the effectiveness of the compression process.
In addition, resource utilization measurements will be collected, including memory consumption, model size, and computational requirements. The compressed model will be compared directly to the original EQTransformer model in order to assess the practical benefits of the proposed approach.
Cross domain testing using the K-NET dataset will provide additional evidence regarding the robustness and generalization capability of the distilled model. The ability to maintain strong performance on unseen seismic data will be considered a key indicator of success.

Evaluation Metrics

The effectiveness of the proposed compression approach will be evaluated using several quantitative metrics. These metrics measure both the predictive performance of the model and its computational efficiency.

Table 2: Evaluation metrics used in the project.

Metric	Description	Purpose in this Project
Precision	Proportion of predicted earthquake detections that correspond to actual events	Measures the reliability of earthquake detection predictions
Recall	Proportion of actual earthquake events successfully detected by the model	Measures the ability to detect seismic events
F1-Score	Harmonic mean of Precision and Recall	Provides an overall measure of detection performance
Accuracy	Proportion of correctly classified samples out of all evaluated samples	Provides an overall measure of earthquake detection performance
MAE	Mean Absolute Error between predicted and ground truth phase arrival times	Measures the accuracy of P-wave and S-wave arrival time predictions
Model Size	Number of parameters and storage requirements	Evaluates compression effectiveness
Inference Time	Average time required to process a single waveform	Measures computational efficiency and real time applicability
Memory Usage	RAM consumption during inference	Evaluates suitability for edge devices

A successful compression strategy should significantly reduce model size, memory consumption, and inference time while maintaining performance metrics close to those of the original EQTransformer model.

AI Tools Used

During the preparation of this project proposal, several Artificial Intelligence tools were used to support literature review, technical understanding, academic writing, and project planning.

- **ChatGPT (OpenAI)**
  - Explain the Earthquake Transformer architecture.
  - Summarize the DistilBERT paper.
  - Compare knowledge distillation approaches.
  - Discuss possible methodological approaches for compressing EQTransformer.
  - Explain the STEAD dataset.
  - Explain the K-NET dataset.
  - Suggest evaluation metrics for model compression.
  - Explain the role of attention mechanisms in Transformer models.
- **Claude (Anthropic)**
  - Review the project structure and identify missing sections.
  - Suggest improvements for the Literature Review.
  - Recommend additional evaluation metrics and validation methods.
  - Review the proposed methodology and identify potential weaknesses.
  - Suggest improvements for the Expected Challenges section.

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Non-Academic Links

1. EQTransformer GitHub Repository:
   https://github.com/smousavi05/EQTransformer
2. DistilBERT Model Documentation:
   https://huggingface.co/distilbert-base-uncased
3. STEAD Dataset Repository:
   https://github.com/smousavi05/STEAD